ECE 280 Lab 2 – TA Solution and Troubleshooting Manual
Signals and Systems: Signal Transformations in Hardware

Use this as a fast grading and troubleshooting guide for:

- Lab 2 manual and report template submissions
- Hardware implementation of time shift, amplitude scaling, and time reversal
- MATLAB verification using measured Arduino-acquired data

1 Quick Pass/Flag Menu

Pass Immediately If All Are True

- Student shows required scope and MATLAB evidence for Steps 1.1 through 3.3.
- Required screenshots include student name/ID and are legible.
- Measured transformations are compared to predicted MATLAB transforms (not only raw captures).
- Report includes NRMSE values for y_1, y_2, y_3, y_4 and discusses largest error source.
- Final conclusions distinguish algorithmic intent from hardware non-idealities.

Flag for Review If Any Are True

- No evidence for one or more transformation steps (2.1–2.4 or 3.1–3.3).
- Shift/reversal claims are visual only, with no quantitative or annotated timing checks.
- MATLAB work appears synthetic-only (no linkage to acquired hardware datasets).
- Same screenshot reused across multiple required steps.

2 Instructor Key: Core Expected Results

Given base signal $x(t)$ and assigned parameters t_0 and B:

$$\begin{aligned} y_1(t) &= x(t - t_0) \\ y_2(t) &= B\,x(t) \\ y_3(t) &= x(-t) \\ y_4(t) &= B\,x(-t - t_0) \end{aligned}$$

Discrete-time mapping (length-N sample vectors):

$y_1[n] \leftrightarrow$ `circshift(x,k)`
$y_2[n] \leftrightarrow$ `B*x`
$y_3[n] \leftrightarrow$ `fliplr(x)`
$y_4[n] \leftrightarrow$ `B*circshift(fliplr(x),k)`

where $k = \lfloor t_0/T_s \rceil$.

TA note: If students use non-periodic windows, circular shift boundary artifacts are expected and should be discussed, not treated as automatic failure.

3 Pre-Lab Solution Guidance

Required Conceptual Targets

- Correct derivations for y_1, y_2, y_3, y_4 from assigned base signal form.
- Correct sign convention for delay/advance and reversal.
- Correct mapping from continuous shift t_0 to integer sample shift k.

Accept/Revise Criteria

- Accept if algebra is correct and plotted/sketched outputs match transformed behavior.
- Revise if student confuses phase shift with amplitude scaling or claims reversal is same as negative amplitude.

4 Measurement Targets and Acceptance Bands

Step 1.1 Base Signal

- Frequency and amplitude should be close to TA-assigned targets.
- Typical acceptance band: within 5–10% if setup is otherwise coherent.

Step 2.1 Amplitude Scaling

- Measured ratio $B_{meas} = A_{Ch1}/A_{Ch2}$ should track assigned B.
- Typical acceptance: ratio error under 10–15% with clear annotation.

Step 2.2 Time Shift

- Cursor-measured delay should match target t_0 within sampling granularity.
- Expected resolution limit: approximately one sample period T_s.

Step 2.3 Time Reversal

- Reversed waveform should preserve amplitude envelope but reverse temporal order.
- Student should identify one distinct feature that appears mirrored in time.

Step 2.4 Combined Transform

- Combined output must simultaneously show reversal pattern, delay offset, and amplitude scaling.

5 Numerical Verification Key (MATLAB)

Expected Deliverables

- Five captured traces: x, y_1, y_2, y_3, y_4.
- Four overlays: measured vs predicted transform for each y_i.
- NRMSE table for all four transforms.

NRMSE Interpretation

$$\text{NRMSE}(\%) = 100 \times \frac{\|y_{meas} - y_{pred}\|_2}{\|y_{pred}\|_2}$$

- Lower NRMSE indicates better hardware-model agreement.
- Time shift and combined transforms commonly show larger error due to alignment and sampling jitter.
- High NRMSE is acceptable if causes are explicitly diagnosed with evidence.

6 Code Key Checks (TA Spot-Check)

Arduino Checks

- Gain variable updated from potentiometer read and applied as scalar multiplier.
- Delay implemented via circular buffer index offset (not by arbitrary blocking delay).
- Reversal implemented via reversed LUT indexing or equivalent index transform.
- Shared variables across ISR/main loop are marked `volatile` if interrupt-timed.

MATLAB Checks

- Predicted transforms generated from captured base x (not hand-crafted replacement arrays).
- Uses correct operators (`circshift`, scalar multiply, `fliplr`).
- Axes labeled and overlay legends clear enough to audit claims.

7 TA Checkoff Script (2–3 Minutes per Team)

1. Ask team to state assigned t_0 and B, then define each y_i verbally.
2. Verify one scope evidence image from Steps 2.1–2.4 with ID visible.
3. Ask for one MATLAB overlay and corresponding NRMSE value.
4. Ask: "Which transform had the largest error and why?"
5. Confirm final report includes all required evidence map items.

8 Troubleshooting Matrix

9 Grading Menu Aligned to Lab 2 Expectations

- Pre-lab derivation quality: formulas, mappings, and sketches are correct.
- Hardware transformation evidence: complete and annotated scope captures.
- MATLAB verification: all required overlays and NRMSE metrics included.
- Technical analysis: strongest error sources identified and justified.
- Authenticity compliance: required screenshots include name/ID.

Table 1: Common Failure Modes and Corrective Actions

Observed Symptom	Likely Cause	TA Action
Amplitude scaling does not change with potentiometer	Pot wired incorrectly or wrong analog pin read	Verify 5V-GND on end terminals and wiper to configured analog input.
Measured delay is inconsistent across runs	Buffer index update bug or nonuniform sampling interval	Check circular-buffer indexing and enforce fixed sample timing.
Time reversal looks like inversion only	Index ordering incorrect; sign flip applied without reversing sample order	Require explicit reverse-index logic and compare with MATLAB `fliplr`.
Combined transform fails while individual steps pass	Wrong operation order (shift before/after reverse mismatch)	Re-derive $y_4 = B\,x(-t - t_0)$ and align firmware/MATLAB operation order.
MATLAB overlays misaligned despite similar shapes	Time vector offset, trigger mismatch, or sample start mismatch	Apply alignment offset/cross-correlation before NRMSE computation.
Very noisy acquired traces	Ground/probe issues or serial acquisition instability	Recheck common ground, probe compensation, and acquisition window consistency.

Suggested Deductions

- Missing required evidence image: -4 to -10, depending on impact.
- No quantitative error metric: -4 to -8.
- Incorrect transform mapping with no correction attempt: -5 to -12.
- Missing ID on required screenshots: -3 to -10.

10 Minimum Passing Submission Standard

A minimally acceptable submission should include:

- Pre-lab derivations for all four transformed signals.
- Scope evidence for base signal and all four hardware transforms.
- MATLAB plots for captured signals and at least one valid measured-vs-predicted overlay.
- NRMSE values for transformation comparisons.
- Brief error analysis linked to measured evidence.